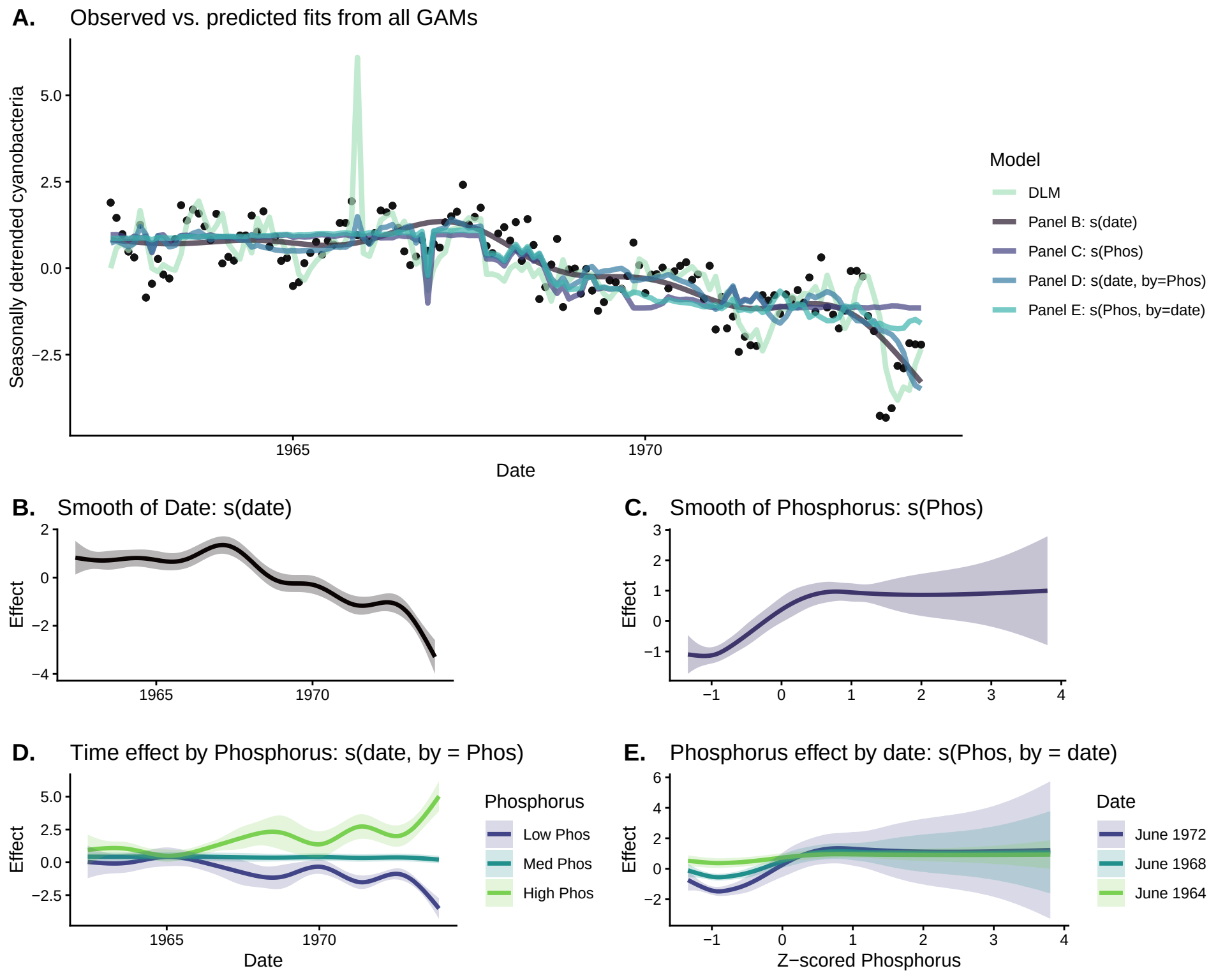
A.
Observed vs. predicted fits from all GAMs
Seasonally detrended cyanobacteria
5.0
2.5
0.0
−2.5
1965
1970
Date
Model
DLM
Panel B: s(date)
Panel C: s(Phos)
Panel D: s(date, by=Phos)
Panel E: s(Phos, by=date)
B.
Smooth of Date: s(date)
Effect
2
0
−2
−4
1965
1970
C.
Smooth of Phosphorus: s(Phos)
Effect
3
2
1
0
−1
−1
0
1
2
3
4
D.
Time effect by Phosphorus: s(date, by = Phos)
Effect
5.0
2.5
0.0
−2.5
1965
1970
Date
Phosphorus
Low Phos
Med Phos
High Phos
E.
Phosphorus effect by date: s(Phos, by = date)
Effect
6
4
2
0
−2
−1
0
1
2
3
4
Z−scored Phosphorus
Date
June 1972
June 1968
June 1964